Worked solutions — class questions and practice

Only consult these solutions after completing the relevant classwork. Check and correct your own reasoning. Homework tasks and homework answers are not included.

Set-up and measurements

1. The sketch should show a shallow, fixed ramp, marked x=0, 0.05, 0.10, 0.15, 0.20 m, and a clearly chosen positive direction.

2. State how the ramp is supported and how the car is consistently released from rest without a push. Times are elapsed from release, not times between marks.

Analysis

3–4. Plot position x in metres against mean elapsed time in seconds. Use sensible labelled axes, units, data points and a smooth best-fit curve.

5. At times corresponding to x=0.05 m and x=0.15 m, draw tangents to the smooth curve. For each tangent choose two readable points, state gradient=rise/run, substitute values and give velocity in m/s.

6. Usually the later tangent is steeper, indicating the car was moving faster later. Base conclusions on your own measurements, not an expected shape.